Shahryar motamedi

astrophysics, galactic dynamics & modified gravity

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Summary

Computational astrophysicist specializing in large-scale numerical modeling and data analysis using Python in Linux-based high-performance computing (HPC) environments, with hands-on experience in using RAMSES (from execution to analysis) as a high resolution open source code for hydrodynamical and gravitational N-body astrophysical simulations, to probe fundamental unsolved questions in galactic dynamics and modified gravity theories.

Online Profiles

GitHub — github.com/shmotamedi
ResearchGate — researchgate.net/profile/Shahryar-Motamedi
LinkedIn — linkedin.com/in/shahryar-motamedi-26b169b7

Education

Ph.D. in Astrophysics 2015–2026
Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran
Advisor: Prof. Hosein Haghi (haghi@iasbs.ac.ir)
M.Sc. in Space Engineering 2010–2013
Sharif University of Technology (SUT), Tehran, Iran
Advisor: Prof. S. H. Pourtakdoust (pourtak@sharif.edu)
B.Sc. in Physics 2006–2010
Sharif University of Technology (SUT), Tehran, Iran

Research Background

Research Assistant IASBS, Zanjan, Iran
Comparing dynamical-friction timescales in Milgromian and Newtonian dynamics
Research Assistant SUT, Tehran, Iran
Investigating chaos in the relativistic restricted three-body problem
Visiting Researcher USTC, Hefei, China (2019)
Developing comparative methods between MOND and DM-based frameworks
Predoctoral Researcher RCISP, Tehran, Iran (2014)
Designing high-precision orbit propagators using analytical perturbation theory

Selected Publication

Motamedi, S., Haghi, H., Ghari, A., Wu, X. and Kroupa, P. (2025). Dynamical friction in Newtonian and Milgromian dynamics. Monthly Notices of the Royal Astronomical Society (MNRAS).

Honors & Awards

Ph.D. Entrance Exam — Ranked first in IASBS entrance exam
M.Sc. Admission — Awarded full scholarship at SUT with direct admission